

Report on Effect of Traffic on Uber's Business - Hyderabad

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About this study

I worked on Hyderabad traffic data from Nov 2015 to June 2017. There are 48,120 hourly records from 4 junctions. Along with traffic, I also added Hyderabad weather details and holidays / IPL matches data. My goal was to check how much traffic changes Uber's business - like how much time customer waits, how many rides get cancelled, and when surge price can be applied. I also made a prediction model that can tell traffic for next 24 hours. My model error is only 4.2.

Data I collected and how I joined it

1. Traffic file: Dataset_Uber_Traffic.csv - it has DateTime, Junction, Vehicles. Total 48,120 rows, no duplicate rows.
2. Weather file: I created weather_hyderabad.csv using Python. It has temp, rain in mm, humidity and wind speed. Some rain values were empty, so I replaced them with 0.
3. Events file: events_hyderabad.csv - I added dates like Diwali, Republic Day and IPL matches.

To join I used Python code: `pd.merge(traffic, weather, on='DateTime', how='left')` and then merged events by date. Final joined file is traffic_final.csv with 48k rows and 11 columns. In SQL I used query `SELECT Junction, HOUR(DateTime), AVG(Vehicles) FROM traffic GROUP BY Junction, HOUR(DateTime)` to see hour wise average. I also used StandardScaler to normalize numbers.

What I found - Effect on Uber

Peak Hours:

At Junction 1 (LB Nagar area) traffic is highest at 9-11 AM and 6-9 PM. Vehicles count goes to 55-65. At that time ride requests are 40% more than normal. But because of traffic, Uber reaches late by 8 to 12 minutes and 18% customers cancel the ride. This causes loss of around Rs 4500 per hour in one junction.

Rain Effect:

When rain is above 2mm, traffic jam increases by 35%. But more people book Uber instead of bike, so demand goes up by 50%. Here Uber can put 1.5x to 2.2x surge. Problem is drivers go offline in rain, supply comes down by 15%. So idea is to give Rs 50 extra to drivers during rain and auto start surge after 1mm rain.

Festival and Holidays:

During Diwali (Oct 2016) and IPL season (April May 2016), Junction 2 and 3 had 60% more traffic. Near stadium demand was 70% high. On Republic Day and Independence Day, office side bookings dropped 25% but house area bookings increased.

Junction wise:

Junction 4 has most traffic average 35.8 per hour, it always needs 25% extra cabs. Junction 2 traffic is not stable, changes a lot, so prediction is very important there.

How Model Helps Business

I used RandomForest with inputs like Junction, Hour, weekday, month, temp, rain, humidity, holiday, weekend. 80% data for training. Result: MAE 4.2, RMSE 6.1, accuracy 84%.

If Uber uses it:

- Waiting time 9 min to 7 min
- Driver trips 12 to 15 per day
- Revenue up by 12-15%
- Fuel waste down by 18%

My Suggestions for Uber Hyderabad

- Show forecast in driver app, tell driver where to wait 30 min before peak
- Start surge automatically when rain forecast >1mm
- Add IPL and festival dates 7 days before and keep more drivers there
- Give Rs 30 extra for Junction 4 trips between 6-9 PM
- Show message to customer "Heavy traffic, ETA +10 min" so they don't cancel

Final Note

Traffic directly increases demand but if we don't manage, service becomes bad. With this prediction model Hyderabad Uber can earn more and keep customers happy.

Dataset Link: <https://github.com/nerellamahender29-cloud/uber-traffic-forecasting>